Sahaj Bhadja

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Summary

Software engineer pursuing an MSCS with 2+ years building **Python infrastructure**, backend tooling, simulation-style evaluation pipelines, CI/CD workflows, and production systems for autonomy-adjacent software.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Advanced NLP, Operating Systems

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **Python-backed infrastructure workflows** for 500+ users, translating requirements into APIs, validation logic, monitored jobs, technical documentation, release notes, and maintainable delivery through reviewed changes and support runbooks.
- Improved **backend reliability and system performance** by tuning indexed queries, background jobs, and CI/CD releases, reducing dashboard latency from **1.2s to 350ms** with production handoffs and diagnostics.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed **Python services and React tooling** handling 10,000+ daily requests, implementing file ingestion, automated checks, documented fixes, and sprint-ready production workflows with code review and monitoring.
- Debugged **production workflow issues** by reproducing OCR, persistence, and execution defects, documenting regression cases, reviewing fixes, and cutting instructor feedback turnaround by **60%** across validation cycles and logs.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **autonomous-driving evaluation pipelines** across 100,000+ trajectory records, producing metrics, logs, and reports for simulation review, model behavior debugging, planning validation, experiment handoffs, and fleet-style analysis.
- Integrated **HD-map geometry and trajectory context** with attention-based encoders, improving planning accuracy by **18%** while surfacing edge cases through closed-loop evaluation, diagnostics, validation reports, and simulation review.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **cross-platform API and ETL services** using Python, SQL, AWS Glue, and S3, improving response time by **30%** across integrations, releases, reviews, handoffs, tooling workflows, and monitoring.
- Maintained **20+ automated test suites** using Jenkins, PyTest, and Mockito across validation flows, increasing CI reliability by **68%** and reducing release regressions during deployments, reviews, and release checks.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built **real-time application modules** with REST APIs, WebSockets, and SQL/NoSQL persistence, supporting 2M+ synchronized records across dashboards, workflows, integrations, reviews, releases, runtime logs, debugging, and validation tooling.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%** while improving deployment consistency, troubleshooting documentation, and shared-environment handoffs across staging workflows and releases.

Skills

Languages: Python, C++, Java, JavaScript, TypeScript, SQL, Bash, C familiarity
Autonomy: simulation pipelines, trajectory data, planning validation, HD maps, offline RL, sensor-data concepts
Backend: FastAPI, Flask, REST APIs, developer tooling, data ingestion, backend services, production debugging
Infra: Docker, Jenkins, CI/CD, GitHub Actions, Linux CLI, AWS, Azure SQL, monitoring, release automation
Data/ML: PyTorch, PostgreSQL, MySQL, MongoDB, SQL Server, metrics, logs, validation, test automation

Projects

CPU Scheduling Simulator Live | *C++, React.js, JavaScript*

- Implemented **FCFS, SJF, SRT, RR, priority, and MLQ scheduling** with metric calculation and visualization, strengthening systems fundamentals, debugging discipline, runtime reasoning, correctness testing, simulation-style analysis, and profiling.

Linguistix Code | *Python, AWS, Docker, CI/CD*

- Developed a **containerized Python REST API** with AWS deployment, CI/CD releases, validation checks, and reproducible execution, achieving **87%** classification accuracy with diagnostic reporting, automated tests, and logs.

Big-Brain Live | *PostgreSQL, GCP, REST APIs, FAISS*

- Built a **cloud data platform** for 50K+ documents with ingestion, metadata persistence, semantic retrieval, and performance tuning, reducing retrieval latency by **40%** through optimized pipelines and indexing.